

Utilizing Geothermal Heat Exchange into Ideal Rankine Power Cycle

Optimized for Desired Net Power Gain and Limiting Exhaustive Environmental Heat Pollution

Rankine Cycles Working Fluid

Ammonia

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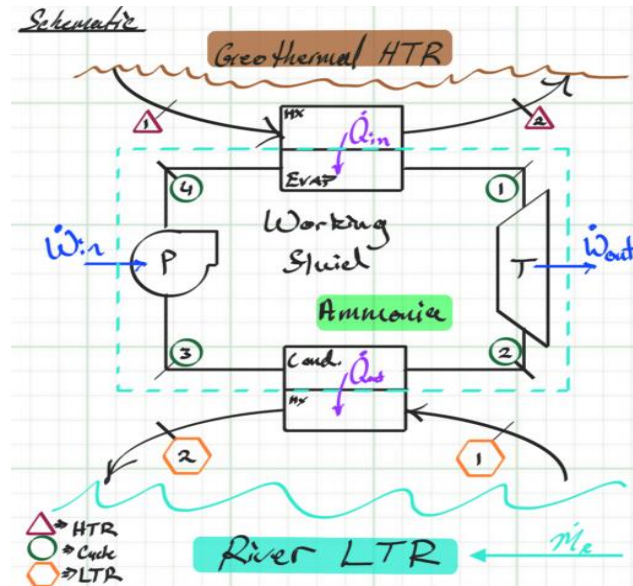
The following report utilizes a geothermal High Temperature Reservoir (HTR) to extract power, while implementing an Ideal Rankine Power Cycle with a working fluid (WF) of Ammonia. My goal is to achieve a net power gain of 5MW optimizing thermal cycle efficiency while limiting environmental exhaustive heat pollution into a “River” (the Low Temperature Reservoir (LTR)) which receives excess thermal energy not bled off by the turbine. I gave respect to linearly varying limit restrictions of the HTR and LTR based on the inlet and outlet pressure, temperature, and mass flow rate at which H₂O interacts with a heat exchanger to evaporate/condense the WF of the power cycle.

To accomplish this goal, I assumed a starting pressure at which the evaporator and condenser operate, this being based on the enthalpy and pressure state points of the WF. Such that the cycle was constrained to have an isentropic process acting across the pump and turbine. As well as having a quality of 1 entering the turbine and quality of 0 entering the pump. These state points were initially picked to fall between 10 and 0.7 MPa on a P-h diagram (optimized to (10.25, 0.675) MPa respectively in EES) of the WF, which fall approximately 5 °c above the max. temp. of the LTR limit out of 11°C and below the min. temp. of 130°C to insure proper heat rate through cycle. I Optimized the cycle based on maximizing the thermal cycle efficiency ($\eta_{cycle,achieved} = 21.63\%$) and minimizing the temp reentering the river ($T_{LTRout,achieved} = 9.249^{\circ}c$).

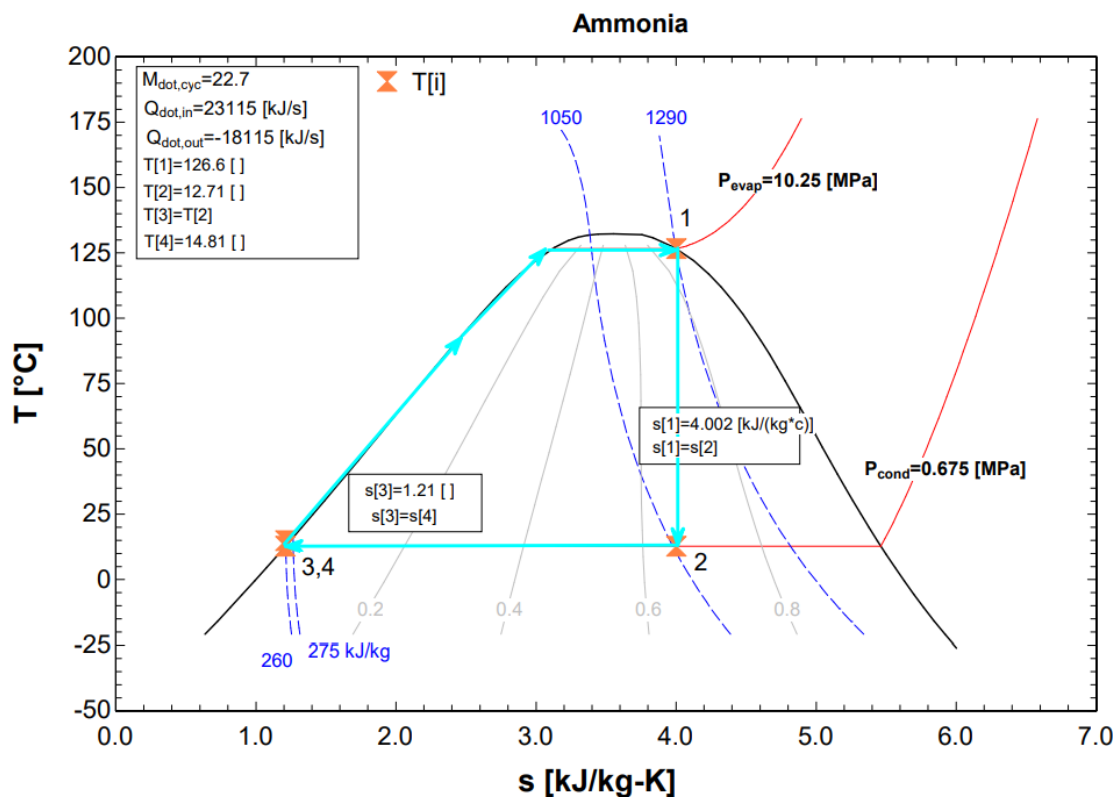
Appendix A

(Power Cycle Design present in the following pages)

i. Schematic



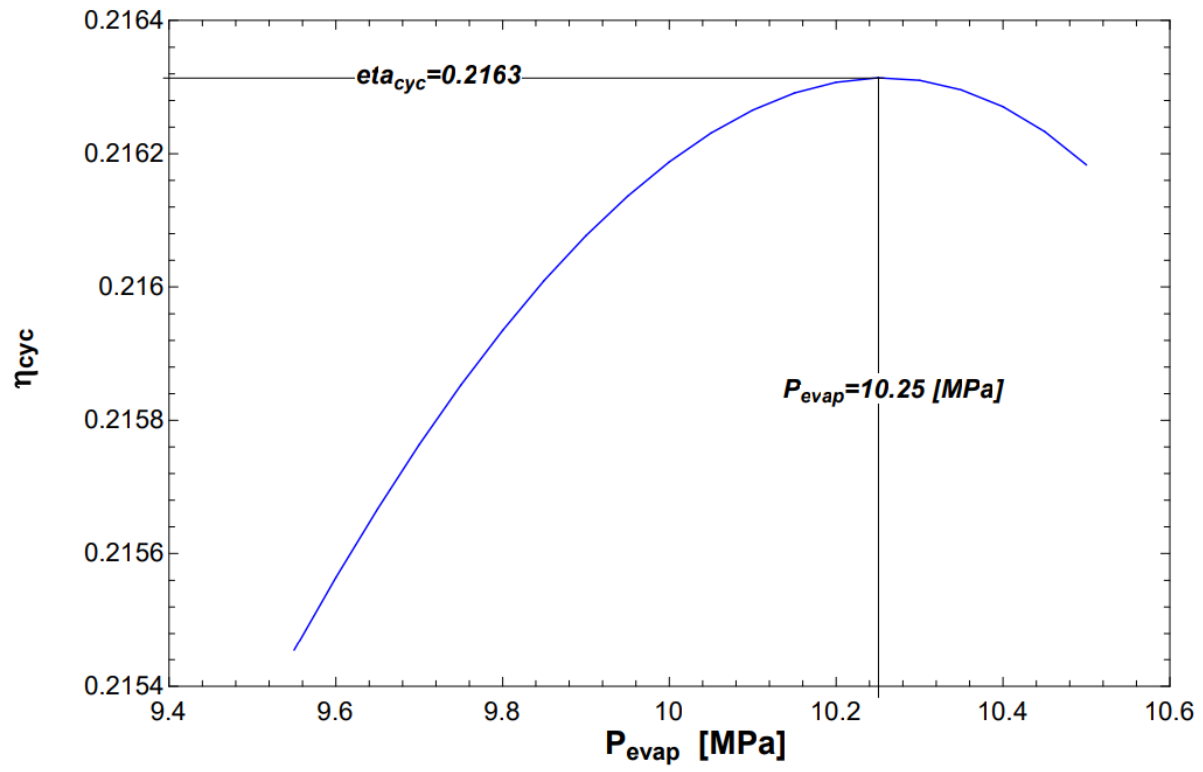
ii. T-s diagram of Ideal Rankine Cycle for Ammonia @ optimal configurations given limit restrictions.



iii. Cycle Operating Specifications

- Working Fluid:
 - Ammonia
- Evaporator pressure:
 - 10250 kPa
- Condenser pressure:
 - 675 kPa
- Cycle Mass flow rate:
 - 22.7 kg/s
- Thermal Cycle Efficiency:
 - 21.63 %
- Geothermal Mass flow rate into Evaporator/Heat Exchanger:
 - 440 kg/s
- Geothermal Reinjection Temperature max limit @ Geothermal Mass flow rate:
 - 144°C
- Geothermal Stream Temperature leaving Evaporator/Heat Exchanger:
 - 137.8 °C
- Max UA limit of Evaporator:
 - 385 kJ/s-°C
- Designed UA for Evaporator:
 - 385 kJ/s-°C
- River Mass flow rate into Condenser/Heat Exchanger:
 - 1015 kg/s
- River Reinjection Temperature max limit @ River Mass flow rate:
 - 9.65 °C
- Temperature of Rejected Stream leaving Condenser/Heat Exchanger:
 - 9.249 °C
- Max UA limit of Condenser:
 - 3500 kJ/s-°C
- Designed UA for Condenser:
 - 3414 kJ/s-°C

iv. η_{cyc} vs P_{evap}



Appendix B

(Written Analysis presented in the following pages)

Knowns

$$P_{\Delta} = 1 \text{ MPa} \quad \dot{M}_R = 140,000 \text{ kg/s}$$

$$T_{\Delta} = 150^{\circ}\text{C} \quad T_R = 5^{\circ}\text{C}$$

$$T_{\Delta \text{ min}} = 130^{\circ}\text{C} \mid \dot{M}_{\Delta} = 300 \text{ kg/s}$$

$$T_{\Delta \text{ max}} = 140^{\circ}\text{C} \mid \dot{M}_{\Delta} = 500 \text{ kg/s}$$

$$\dot{M}_{\Delta} = 500 \text{ kg/s} \Rightarrow \text{Absolute max}$$

$$P_{\Delta} = 0.11 \text{ MPa}$$

$$\dot{M}_{\Delta \text{ max}} = 1\% \dot{M}_R$$

$$\Rightarrow \dot{M}_{\Delta \text{ max}} = 1400 \text{ kg/s}$$

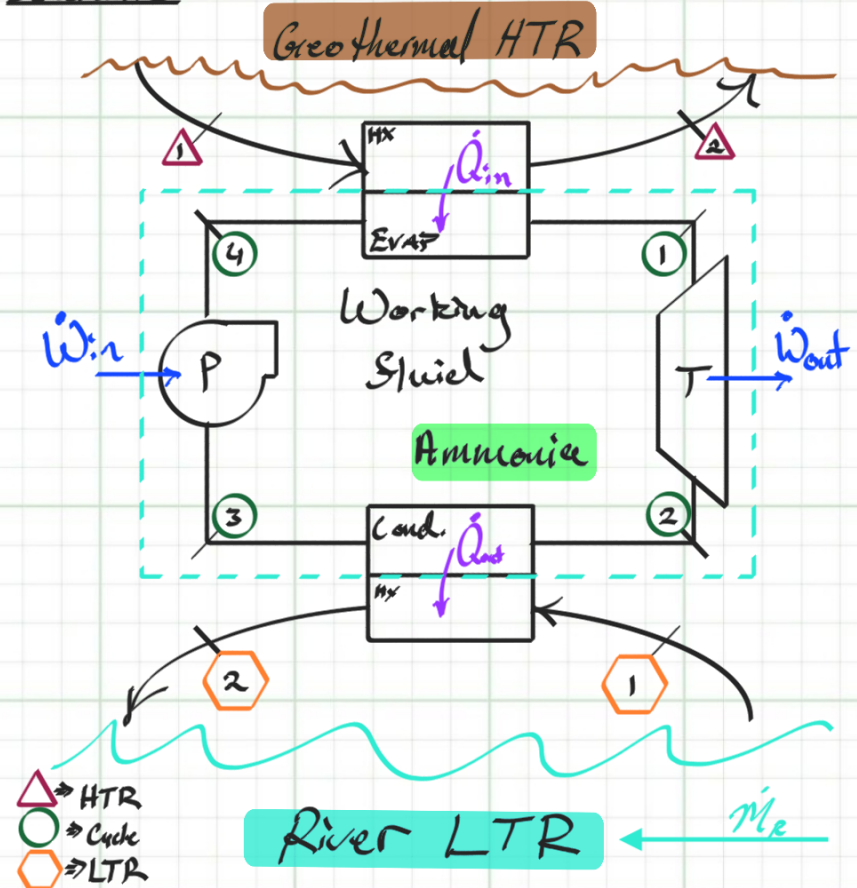
$$T_{\Delta \text{ min}} = 8^{\circ}\text{C} \mid \dot{M}_{\Delta} = 1\% \dot{M}_R$$

$$T_{\Delta \text{ max}} = 11^{\circ}\text{C} \mid \dot{M}_{\Delta} = 0.5\% \dot{M}_R$$

$$U_{A \text{ max, evap}} = 385 \text{ W/m}^2$$

$$U_{A \text{ max, cond}} = 3500 \text{ W/m}^2$$

Schematic



Determine *Based on Ammonia*

1) Symbolic 1st Law of Comp. in I.D. stand. Rankine $\Rightarrow$ Generate Parametric table

2) P_{Δ} (Isobaric Pressure across Condenser) $\left. \begin{array}{l} 1^{\text{st}} \text{ based on P-h diag} \\ 2^{\text{nd}} \text{ thru EES} \end{array} \right\}$
 P_{Δ} (Isobaric Pressure across Evaporator)

3) $\dot{M}_{\text{cycle}} \mid W_{\text{net}} = 5 \text{ MW}$ *Note: important to find $\dot{Q}_{\text{net}}$ in (kW)*

4) Generate Parametric table w/ P_{Δ} , $\dot{M}_{\text{cycle}}$

Using 50 kPa Increments vary P_{Δ} to see influence on

$\dot{M}_{\text{cycle}}$ to Determine optimal $\dot{M}_{\text{cycle}} \mid U_{A \text{ evap}} \neq U_{A \text{ evap, max}}$

* Note: name Table 1 *

5) Plot $\dot{M}_{\text{cycle}}$ vs. P_{Δ}

6) Manually vary P_{Δ} in 5 kPa increments to determine

$$P_{\Delta \text{ min}} \left\{ \begin{array}{l} U_{A \text{ tot, cycle}} \neq 0 \\ U_{A \text{ cond}} \neq U_{A \text{ cond, max}} \end{array} \right.$$

- Including $U_{A \text{ evap}}$, T_{Δ} , $T_{\Delta \text{ max}}$

Vary $\dot{M}_{\Delta}$ between known limits in 10 kg/s increments then determine

optimal $\dot{M}_{\Delta} \mid T_{\Delta} \neq T_{\Delta \text{ min}}$
 $U_{A \text{ evap}} \neq U_{A \text{ evap, max}}$

* Note: name Table 2 *

8) Generate Parametric table

- Including $U_{A \text{ evap}}$, T_{Δ} , $T_{\Delta \text{ max}}$

Vary $\dot{M}_{\Delta}$ between known limits in 10 kg/s increments then determine

optimal $\dot{M}_{\Delta} \mid T_{\Delta \text{ min}} \leq T_{\Delta} \leq T_{\Delta \text{ max}}$
 $U_{A \text{ cond}} \leq U_{A \text{ cond, max}}$

* Note: name Table 3 *

Symbolic

C.O.E. * For I.D Rankine *

$$\text{Turbine: } 0 = \dot{Q}_1 - \dot{W}_1 + \dot{M}_{\text{cyc}}(h_1 - h_2)$$

$$\Rightarrow \dot{W}_1 = \dot{M}_{\text{cyc}}(h_1 - h_2)$$

$$\text{Pump: } 0 = \dot{Q}_2 - \dot{W}_2 + \dot{M}_{\text{cyc}}(h_2 - h_3)$$

$$\Rightarrow \dot{W}_2 = \dot{M}_{\text{cyc}}(h_3 - h_2)$$

$$\text{Evap: } 0 = \dot{Q}_3 - \dot{W}_3 + \dot{M}_{\text{cyc}}(h_3 - h_4)$$

$$\Rightarrow \dot{Q}_3 = \dot{M}_{\text{cyc}}(h_4 - h_3)$$

$$\text{Cond: } 0 = \dot{Q}_4 - \dot{W}_4 + \dot{M}_{\text{cyc}}(h_4 - h_1)$$

$$\Rightarrow \dot{Q}_4 = \dot{M}_{\text{cyc}}(h_1 - h_4)$$

* Geo thermal HTR *

$$0 = \dot{Q}_A - \dot{W}_A + \dot{M}_A(h_A - h_B)$$

$$\Rightarrow \dot{Q}_A = \dot{M}_A(h_A - h_B)$$

* River LTR *

$$0 = \dot{Q}_B - \dot{W}_B + \dot{M}_B(h_B - h_C)$$

$$\Rightarrow \dot{Q}_B = \dot{M}_B(h_B - h_C)$$

Assumptions for I.D conditions

- Mechanical Components are Adiabatic
- Non-Mechanical Components no boundary work is done
- Neglect $\Delta PE, \Delta KE$
- S.S. open sys for individual components
- S.S. closed sys for individual cycles.
- Geo thermal HTR
- River LTR
- Rankine Cycle
- $\eta_t + \eta_p = 1$

$$\dot{Q}_A = \dot{Q}_1 = \dot{Q}_{\text{in}}$$

$$\dot{Q}_B = \dot{Q}_4 = \dot{Q}_{\text{out}}$$

$$\dot{W}_{\text{net}} = \dot{W}_1 + \dot{W}_2$$

$$\eta_{\text{cyc}} = \frac{\dot{W}_{\text{net}}}{\dot{Q}_{\text{in}}}$$

$$|\dot{W}_{\text{net}} \dot{M}_{\text{cyc}}| = \dot{W}_{\text{net}}$$

* Note: $\dot{M}_{\text{cyc}} = \text{constant}$ *

$$UA_{\text{EVAP}} = \frac{481 \dot{M}_{\text{cyc}}}{\Delta T_{\text{LME}}}$$

$$\Delta T_{\text{LME}} = \frac{\Delta T_{e1} - \Delta T_{e2}}{\ln\left(\frac{\Delta T_{e1}}{\Delta T_{e2}}\right)}$$

$$\Delta T_{e1} = T_A - T_1$$

$$\Delta T_{e2} = T_A - T_2$$

Equations given

$$UA_{\text{COND}} = \frac{1282 \dot{M}_{\text{cyc}}}{\Delta T_{\text{LMC}}}$$

$$\Delta T_{\text{LMC}} = \frac{\Delta T_{c1} - \Delta T_{c2}}{\ln\left(\frac{\Delta T_{c1}}{\Delta T_{c2}}\right)}$$

$$\Delta T_{c1} = T_1 - T_2$$

$$\Delta T_{c2} = T_2 - T_3$$

* Note: To find T_1 & T_2 initially set $T_1 = T_{\text{sat, vap}} 5^\circ\text{C}$ lower than T_A then hold
Then set $T_2 = T_{\text{sat, liq}} 5^\circ\text{C}$ higher than T_1 then hold. Refer to P-h diagram below to infer P values
The value of h can be inferred under Assumed / known conditions which
will determine $\dot{M}_{\text{cyc}}$. T_2 and T_A will vary as $\dot{M}_B$ and $\dot{M}_A$ linearly vary must find
 T_2 and T_A such that all conditions are satisfied.

State Points found In CES on Known Intensive Properties

State Points | 2 Ind. Intensive Prop.

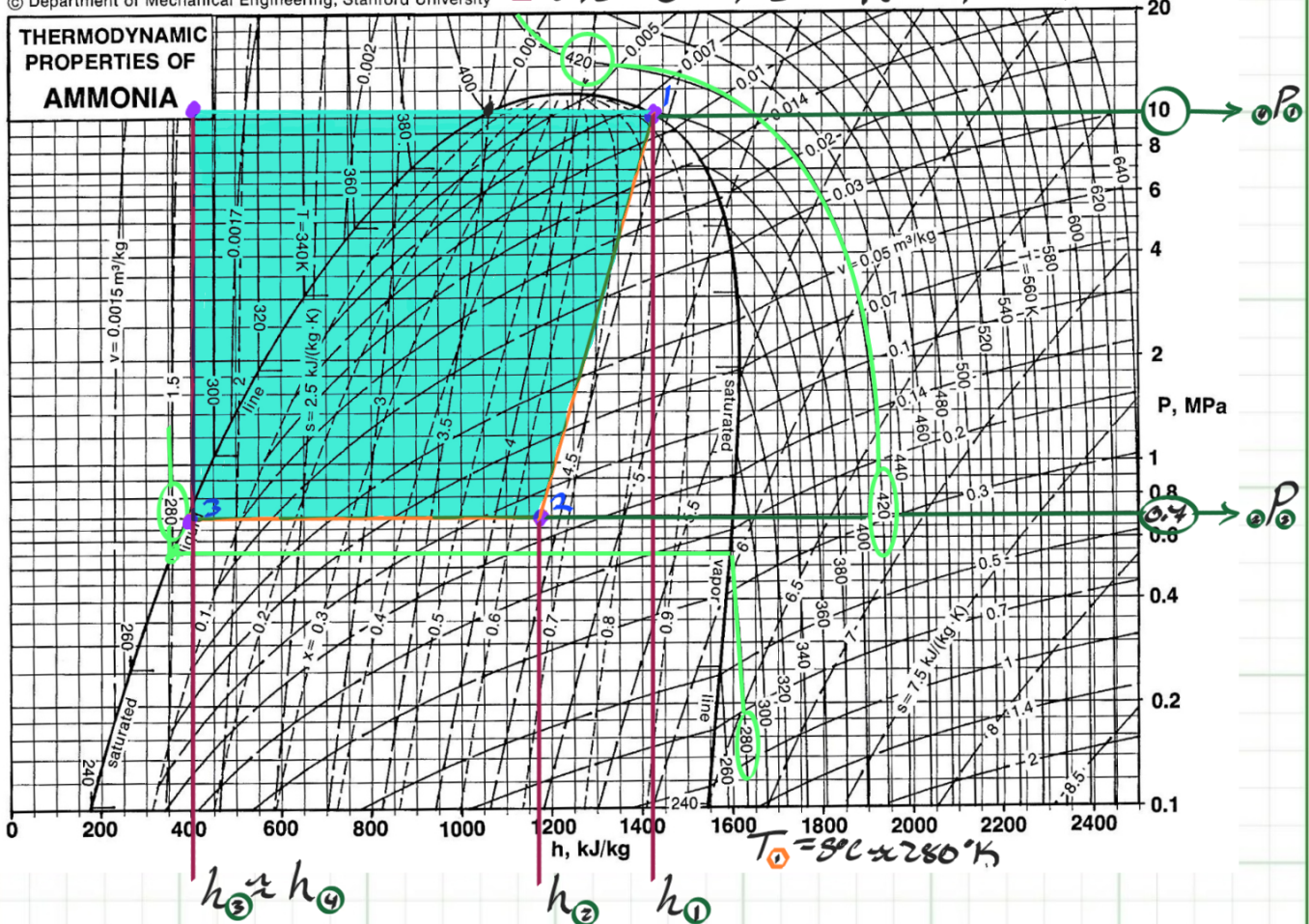
1	$P = P_1, T = T_1$
2	$P = P_1, T_{\min} \leq T \leq T_{\max}$
1	$X = 1, P = P_0$
2	$S = S_1 = S_2, P = P_0$
3	$x = 0, P = P_0$
4	$S = S_3 = S_4, P = P_0$
1	$P = P_0, T = T_0$
2	$P = P_0, T_{\min} \leq T \leq T_{\max}$

* T is dependent on linear *
Variation of ΔM_1

* T is dependent on linear *
Variation of ΔM_2

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$$T_{\Delta} \approx 150^\circ\text{C} = 423.15^\circ\text{K} \approx 423^\circ\text{K}$$



$$h_1 \approx 1420 \text{ kJ/kg} \quad h_3 \approx 400 \text{ kJ/kg}$$

$$h_2 \approx 1170 \text{ kJ/kg} \quad h_4 \approx 400 \text{ kJ/kg}$$

Appendix C

(EES code and outputs for Design presented in the following pages)

{Knowns}

P_HTR=1 [MPa]

T_HTR_in=150 [c]

P_LTR=0.11[MPa]

T_LTR_in= 5 [c]

UA_evap_max=385 [kJ/(s*c)]

UA_cond_max=3500[kJ/(s*c)]

*{Initially set to Limit values desired (i.e. T_HTR_out_min_limit=130[c], M_dot_HTR=300[kg/s], T_LTR_out_limit_max=8[c], M_dot_LTR=1400[kg/s]) Alter to optimal temperature \ mas flow rate as tables are generated}**{comment out when generating Table2, change # to the desired corresponding value from table, closet to Ua_evap_max. Uncomment to just solve for varribles. This value is utilized to calculate the appropriate Cp value as it will vary as values are updated }*

T_HTR_out_max_limit=137.8 [c]

{comment out when generating Table2, change # to the desired corresponding value from table, closet to UA_evap_max. Uncomment to just solve for varribles}

M_dot_HTR=440[kg/s]

{comment out when generating Table3, change # to the desired corresponding value from table, closet to UA_cond_max. Uncomment to just solve for varribles. This value is utilized to calculate the appropriate Cp value as it will vary as values are updated}

T_LTR_out_limit_max=9.249 [c]

{comment out when Generating Table3, change # to the desired corresponding value from table, closet to UA_cond_max. Uncomment to just solve for varribles}

M_dot_LTR=1015[kg/s]

*{Chosen based on P-h diagram initially but will change depeding on table infromation}**P_evap=10.25[MPa] {comment out when generating Table1, change # to the desired corresponding value from table.Uncomment to just solve for varribles}*

P_cond= 0.675 [MPa]

{thermal Cycle Efficiency} $\eta_{cyc} = \text{abs}(W_{dot_net}/Q_{dot_in})$ *{Desired Net Work of Cycle}*

W_dot_net=5000 [kJ/s]

{relationship of net work to determine Mass flow rate of cycle} $w_{net} = w_{1_2} + w_{3_4}$ $W_{dot_net} = M_{dot_cyc} * w_{net}$ *{relationship of Qin to Mass flow rate of cycle}* $Q_{dot_in} = q_{4_1} * M_{dot_cyc}$ $Q_{dot_out} = q_{2_3} * M_{dot_cyc}$ *{C.O.E. cycle components}*

$$w_{1_2} = h[1] - h[2]$$

$$w_{3_4} = h[3] - h[4]$$

$$q_{4_1} = h[1] - h[4]$$

$$q_{2_3} = h[3] - h[2]$$

{Cycle State Points based on I.D Standard rankine}

$$h[1] = \text{enthalpy}(\text{Ammonia}, P=P_{\text{evap}}, x=1)$$

$$s[1] = \text{entropy}(\text{Ammonia}, P=P_{\text{evap}}, x=1)$$

$$T[1] = \text{temperature}(\text{Ammonia}, P=P_{\text{evap}}, x=1)$$

$$h[2] = \text{enthalpy}(\text{Ammonia}, P=P_{\text{cond}}, s=s[1])$$

$$T[2] = \text{temperature}(\text{Ammonia}, P=P_{\text{cond}}, s=s[2])$$

$$s[2] = s[1]$$

$$h[3] = \text{enthalpy}(\text{Ammonia}, P=P_{\text{cond}}, x=0)$$

$$s[3] = \text{entropy}(\text{Ammonia}, P=P_{\text{cond}}, x=0)$$

$$T[3] = \text{temperature}(\text{Ammonia}, P=P_{\text{cond}}, x=0)$$

$$h[4] = \text{enthalpy}(\text{Ammonia}, P=P_{\text{evap}}, s=s[4])$$

$$T[4] = \text{temperature}(\text{Ammonia}, P=P_{\text{evap}}, s=s[4])$$

$$s[4] = s[3]$$

{Temp Limit averages for Cp of HTR and LTR}

$$T_{\text{HTR_limit_avg}} = (T_{\text{HTR_in}} + T_{\text{HTR_out_max_limit}}) / 2$$

$$T_{\text{LTR_limit_avg}} = (T_{\text{LTR_in}} + T_{\text{LTR_out_limit_max}}) / 2$$

{Given Equations based on UA of Cond and Evap}

$$UA_{\text{evap}} = (Q_{\text{dot_in}}) / T_{\text{limit_evap}}$$

$$T_{\text{limit_evap}} = (\Delta T_{\text{evap_1}} - \Delta T_{\text{evap_2}}) / \ln(\Delta T_{\text{evap_1}} / \Delta T_{\text{evap_2}})$$

$$\Delta T_{\text{evap_1}} = T_{\text{HTR_in}} - T[1]$$

$$\Delta T_{\text{evap_2}} = T_{\text{HTR_out}} - T[4]$$

$$UA_{\text{cond}} = (Q_{\text{dot_out}}) / T_{\text{limit_cond}}$$

$$T_{\text{limit_cond}} = (\Delta T_{\text{cond_1}} - \Delta T_{\text{cond_2}}) / \ln(\Delta T_{\text{cond_1}} / \Delta T_{\text{cond_2}})$$

$$\Delta T_{\text{cond_1}} = T_{\text{LTR_in}} - T[3]$$

$$\Delta T_{\text{cond_2}} = T_{\text{LTR_out}} - T[2]$$

{Relation between Q rate out of HTR and into Cycle}

$$Q_{\text{dot_HTR_cyc}} = -Q_{\text{dot_in}}$$

$$Q_{\text{dot_HTR_cyc}} = M_{\text{dot_HTR}} \cdot c_{p_H2o_HTR} \cdot (T_{\text{HTR_out}} - T_{\text{HTR_in}})$$

$$c_{p_H2o_HTR} = c_p(\text{Steam}, T=T_{\text{HTR_limit_avg}}, P=P_{\text{HTR}}) \quad \{Cp_{h2O} \text{ approximated based off HTR Temp Limits Cp change is negligible due to changes in out temp as it changes across the Temp Limit Range}\}$$

{Relation between Q rate out of LTR and into Cycle}

$$Q_{\text{dot_LTR_cyc}} = \text{abs}(Q_{\text{dot_out}})$$

$$Q_{\text{dot_LTR_cyc}} = M_{\text{dot_LTR}} \cdot c_{p_H2o_LTR} \cdot (T_{\text{LTR_out}} - T_{\text{LTR_in}})$$

$$c_{p_H2o_LTR} = c_p(\text{Steam}, T=T_{\text{LTR_limit_avg}}, P=P_{\text{LTR}})$$

$$\{Cp_{h2O} \text{ approximated based off LTR Temp Limits Cp change is negligible due to changes in out temp as it changes across the Temp Limit Range}\}$$

SOLUTION

Unit Settings: SI C MPa kJ mass deg

$$c_{p_{H2O,HTR}} = 4.29 \quad [\text{kJ/kg} \cdot \text{C}]$$

$$\delta T_{\text{cond},1} = -7.711 \quad [\text{C}]$$

$$\delta T_{\text{evap},1} = 23.43 \quad [\text{C}]$$

$$c_{p_{H2O,LTR}} = 4.2 \quad [\text{kJ/kg} \cdot \text{C}]$$

$$\delta T_{\text{cond},2} = -3.462 \quad [\text{C}]$$

$$\delta T_{\text{evap},2} = 122.9 \quad [\text{C}]$$

$\eta_{cyc} = 0.2163$
 $\dot{M}_{HTR} = 440 \text{ [kg/s]}$
 $P_{cond} = 0.675 \text{ [MPa]}$
 $P_{HTR} = 1 \text{ [MPa]}$
 $q_{2,3} = -798.1 \text{ [kJ/kg]}$
 $\dot{Q}_{HTR,cyc} = -23115 \text{ [kJ/s]}$
 $\dot{Q}_{LTR,cyc} = 18115 \text{ [kJ/s]}$
 $T_{HTR,in} = 150 \text{ [C]}$
 $T_{HTR,out} = 137.8 \text{ [C]}$
 $T_{limit,cond} = -5.306 \text{ [C]}$
 $T_{LTR,in} = 5 \text{ [C]}$
 $T_{LTR,out} = 9.249 \text{ [C]}$
 $UA_{cond} = 3414 \text{ [kJ/(s}^{\circ}\text{C)]}$
 $UA_{evap} = 385 \text{ [kJ/(s}^{\circ}\text{C)]}$
 $w_{1,2} = 235.6 \text{ [kJ/kg]}$
 $\dot{W}_{net} = 5000 \text{ [kJ/s]}$

$\dot{M}_{cyc} = 22.7 \text{ [kg/s]}$
 $\dot{M}_{LTR} = 1015 \text{ [kg/s]}$
 $P_{evap} = 10.25 \text{ [MPa]}$
 $P_{LTR} = 0.11 \text{ [MPa]}$
 $q_{4,1} = 1018 \text{ [kJ/kg]}$
 $\dot{Q}_{in} = 23115 \text{ [kJ/s]}$
 $\dot{Q}_{out} = -18115 \text{ [kJ/s]}$
 $T_{HTR,limit,avg} = 143.9 \text{ [C]}$
 $T_{HTR,out,max,limit} = 137.8 \text{ [C]}$
 $T_{limit,evap} = 60.03 \text{ [C]}$
 $T_{LTR,limit,avg} = 7.125 \text{ [C]}$
 $T_{LTR,out,limit,max} = 9.249 \text{ [C]}$
 $UA_{cond,max} = 3500 \text{ [kJ/(s}^{\circ}\text{C)]}$
 $UA_{evap,max} = 385 \text{ [kJ/(s}^{\circ}\text{C)]}$
 $w_{3,4} = -15.37 \text{ [kJ/kg]}$
 $W_{net} = 220.3 \text{ [kJ/kg]}$

No unit problems were detected.

Arrays Table: Main

	T_i [C]	s_i [kJ/(kg $^{\circ}$ C)]	h_i [kJ/kg]
1	126.6	4.002	1293
2	12.71	4.002	1057
3	12.71	1.21	259.1
4	14.81	1.21	274.5

Parametric Table: Table 1

	P_{evap} [MPa]	$\dot{M}_{cyc}$ [kg/s]	UA_{evap} [kJ/(s $^{\circ}$ C)]	η_{cyc}
Run 1	10.5	23.1	394	0.2162
Run 2	10.45	23.01	392.1	0.2162
Run 3	10.4	22.93	390.3	0.2163
Run 4	10.35	22.85	388.5	0.2163
Run 5	10.3	22.77	386.7	0.2163
Run 6	10.25	22.7	385	0.2163
Run 7	10.2	22.63	383.4	0.2163
Run 8	10.15	22.56	381.7	0.2163
Run 9	10.1	22.5	380.1	0.2163
Run 10	10.05	22.44	378.5	0.2162
Run 11	10	22.38	377	0.2162
Run 12	9.95	22.32	375.5	0.2161
Run 13	9.9	22.27	374	0.2161
Run 14	9.85	22.22	372.6	0.216
Run 15	9.8	22.17	371.1	0.2159
Run 16	9.75	22.12	369.7	0.2159
Run 17	9.7	22.08	368.3	0.2158
Run 18	9.65	22.03	367	0.2157
Run 19	9.6	21.99	365.7	0.2156
Run 20	9.55	21.95	364.4	0.2155

Parametric Table: Table 2

	$\dot{M}_{\text{HTR}}$ [kg/s]	$T_{\text{HTR,out}}$ [c]	$T_{\text{HTR,out,max,limit}}$ [C]	UA_{evap} [kJ/(s*c)]
Run 1	500	139.3	150	382.1
Run 2	490	139	149	382.5
Run 3	480	138.8	148	383
Run 4	470	138.6	147	383.4
Run 5	460	138.3	146	383.9
Run 6	450	138.1	145	384.4
Run 7	440	137.8	144	385
Run 8	430	137.5	143	385.6
Run 9	420	137.2	142	386.2
Run 10	410	136.9	141	386.8
Run 11	400	136.5	140	387.5
Run 12	390	136.2	139	388.2
Run 13	380	135.8	138	388.9
Run 14	370	135.4	137	389.7
Run 15	360	135	136	390.5
Run 16	350	134.6	135	391.4
Run 17	340	134.1	134	392.4
Run 18	330	133.7	133	393.4
Run 19	320	133.1	132	394.5
Run 20	310	132.6	131	395.6
Run 21	300	132	130	396.8

Parametric Table: Table 3

	$\dot{M}_{\text{LTR}}$ [kg/s]	$T_{\text{LTR,out}}$ [c]	$T_{\text{LTR,out,limit,max}}$ [C]	UA_{cond} [kJ/(s*c)]
Run 1	1400	8.08	8	2999
Run 2	1365	8.159	8.15	3022
Run 3	1330	8.242	8.3	3048
Run 4	1295	8.33	8.45	3075
Run 5	1260	8.422	8.6	3105
Run 6	1225	8.52	8.75	3138
Run 7	1190	8.624	8.9	3173
Run 8	1155	8.734	9.05	3212
Run 9	1120	8.851	9.2	3255
Run 10	1085	8.975	9.35	3302
Run 11	1050	9.108	9.5	3355
Run 12	1015	9.249	9.65	3414
Run 13	980	9.401	9.8	3481
Run 14	945	9.564	9.95	3557
Run 15	910	9.74	10.1	3645
Run 16	875	9.93	10.25	3747
Run 17	840	10.14	10.4	3868
Run 18	805	10.36	10.55	4014
Run 19	770	10.6	10.7	4193
Run 20	735	10.87	10.85	4420
Run 21	700	11.16	11	4720

